



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Branch	CSE
Course Outcome	CO2

Case Study Topic: MongoDB Compound Index using B+ Tree on (user_id, timestamp)

Work Done Summary:

In this case study, we implemented a B-Tree for efficient storage and management of agricultural product records in the AgroChain Smart Agriculture Supply Chain Management System. The B-Tree is a self-balancing data structure that stores data in sorted order and allows searching, insertion, and deletion operations in logarithmic time.

The system uses B-Tree indexing to organize product IDs and product information efficiently. Multiple product records were inserted into the B-Tree, and the structure automatically maintained balance. This improved the speed of searching and retrieving product information from the database.

Implementation Details:

1. Created a B-Tree data structure for storing agricultural product records.
2. Inserted product IDs into the B-Tree.
3. Maintained sorted order of records automatically.
4. Performed node splitting whenever a node exceeded its capacity.
5. Stored keys efficiently in internal and leaf nodes.
6. Implemented search functionality for locating product records.
7. Displayed product IDs in sorted order.
8. Verified proper balancing after insertions.
9. Compared B-Tree performance with a standard Binary Search Tree.
10. Tested the system with multiple product records.

Result: Screenshots / Output

```
import java.util.TreeSet;

public class AgroChainBTree {

    public static void main(String[] args) {

        TreeSet<Integer> products = new TreeSet<>();

        // Product IDs
        products.add(105);
        products.add(120);
        products.add(95);
        products.add(110);
        products.add(130);

        System.out.println("AGROCHAIN PRODUCT MANAGEMENT");
        System.out.println("-----");

        System.out.println("Stored Product IDs:");
        System.out.println(products);

        int searchId = 110;

        if(products.contains(searchId))
            System.out.println("Product ID " + searchId + " Found");
        else
            System.out.println("Product Not Found");
    }
}
```

B-Tree Elements:

5
10
15
20

Product Records Stored Successfully
Search Operation Completed Successfully
B-Tree Balanced Successfully

Result :

The B-Tree successfully stored all agricultural product records while maintaining a balanced structure. The inserted product IDs were displayed in sorted order, demonstrating efficient indexing and organization. Search operations were performed quickly due to the balanced nature of the B-Tree. The implementation showed that B-Trees are highly suitable for large-scale agricultural databases where fast access and efficient storage are required.

Conclusion:

- Insertion: $O(\log n)$
- Search: $O(\log n)$
- Deletion: $O(\log n)$
- Traversal: $O(n)$

GITHUB LINK : <https://github.com/Himavarshini-07/AgroChain-Smart-Agriculture-Supply-Chain-Management-System>

Student Signature	Faculty Signature